

A Breathing-Based Meditation Intervention for Patients With Major Depressive Disorder Following Inadequate Response to Antidepressants: A Randomized Pilot Study

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J Clin Psychiatry 2017; 78(1):e59-e68 | DOI: 10.4088/JCP.16m10819 | PMCID: PMC5272872

ABSTRACT

Objective: To evaluate the feasibility, efficacy, and tolerability of a breathing-based program (Sudarshan Kriya Yoga, SKY) as an adjunctive treatment in outpatients with major depressive disorder (MDD) who had an inadequate response to continuous antidepressant therapy.

Method: 8-week randomized controlled clinical pilot study was conducted from May to March 2015. Depressed patients (HDRS-17 score ≥ 14 despite ≥ 8 weeks of antidepressant) were randomized to either SKY or a waitlist control group. The primary outcome measure was in 17-item Hamilton Depression Rating Scale (HDRS-17) total score from baseline to week 8. Secondary outcomes included the Beck Depression Inventory (BDI) and Beck Anxiety Inventory (BAI).

Results: Fifty patients with MDD were enrolled and randomized (SKY: n=13; waitlist: n=12). The SKY intervention produced a significant reduction in depression and anxiety scores compared to the waitlist control. The change in HDRS-17 score in the intent-to-treat SKY group was -9.77 ± 1.90 compared to $+0.50$ (SE = 1.87) in waitlist control (difference = 10.27 points, $p = 0.0032$). Mean BDI score decreased by 17.23 points in SKY versus 1.75 in controls ($p = 0.0101$). Mean BAI score decreased by 5.44 points in SKY versus 0.25 in controls ($p = 0.0097$). Remission rates in the completed SKY group reached 36.4% on HDRS-17.

Conclusions: Sudarshan Kriya yogic breathing is a feasible, well-tolerated, and effective intervention for patients with major depressive disorder who have had an inadequate response to antidepressant pharmacotherapy.

CLINICAL POINTS

- More than 50% of patients with MDD fail to achieve full remission with first-line antidepressant therapy.
- Sudarshan Kriya Yoga provides a safe, low-cost, non-pharmacological adjunctive treatment that significantly reduces both depressive symptoms and co-occurring anxiety without medication-related adverse side effects.

PRIMARY & SECONDARY CLINICAL OUTCOME MEASURES (WEEK 8 RCT DATA)

Clinical Rating Scale	Waitlist Control (n=12)	SKY Group (n=13)	Difference / Impact	P-Value
17-Item Hamilton Depression (HDRS)	+0.50 (1.87)	-9.77 (1.90)	-10.27 pts drop	p = 0.0032
Beck Depression Inventory (BDI-II)	-1.75 (2.56)	-17.23 (2.70)	-15.48 pts drop	p = 0.0101
Beck Anxiety Inventory (BAI)	-0.25 (1.48)	-5.44 (1.60)	-5.19 pts drop	p = 0.0097
HDRS Response Rate (>50% Drop)	8.3%	54.5% (Completers)	+46.2% Gain	p = 0.0271
Clinical Remission (HDRS ≤ 7)	8.3%	36.4% (Completers)	+28.1% Gain	p = 0.1550

NEUROBIOLOGICAL MECHANISMS & DISCUSSION

Sudarshan Kriya Yoga involves cyclical, rhythmic breathing patterns that directly stimulate pulmonary stretch receptors and vagal afferents. Afferent vagal signals project to the nucleus tractus solitarius and locus coeruleus, modulating thalamo-cortical loops and down-regulating hyperactive amygdala circuits. Additionally, SKY has been demonstrated in concurrent biomarker trials to decrease serum cortisol, reduce inflammatory cytokines (TNF-alpha, IL-6), and upregulate Brain-Derived Neurotrophic Factor (BDNF), providing a neurobiological mechanism for cellular repair and emotional resilience in treatment-resistant depressive states.